Chapter 13

The Use of Surfactants to Enhance Particle Removal from Surfaces

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Chapter Outline

13.1 INTRODUCTION

13.1.1 Industrial Perspective

Many of the items we use daily must be manufactured in environments that require particulate materials, yet the particles that are essential to production must be removed to extremely low levels following the relevant manufacturing processes. A single particle remaining at a critical place on a semiconductor circuit during the manufacturing process can cause circuit failure in a finished

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integrated circuit. Consequently, particle removal techniques are vital to electronic circuit manufacturing.[1–18] Other industries such as optical component manufacturing also rely on particle removal techniques to create quality components with appropriate finishes.

In many industries, particles are handled in aqueous media, and particle removal from surfaces is also performed in aqueous media. Removal techniques vary from simple brush scrubbing techniques[19–22] and megasonic vibration[23–26] for wet surfaces to laser treatment[27–30] and water ice[31] and CO_2 snow[32] or air[33] cleaning for dry surfaces.

Particles adhere to surfaces due to natural attractive forces between particles and surfaces. Surfactants can effectively reduce natural attractive forces between particles and surfaces, thereby enhancing particle removal. This chapter discusses the use of surfactants in enhancing particle removal from surfaces in aqueous environments, although the discussion presented here can also be applied to other environments. Related information is also found in other literature sources.[34,35]

13.1.2 Historical Perspective

Powders have been used for grinding and polishing of jewelry and ornaments for millennia, yet an understanding of how particles interact with surfaces was relatively obscure until the 1940s.[36] Most of the present knowledge about particle-surface interactions can be related to research that developed in the 1940s. However, direct, accurate measurement of such interaction forces between fine particles and surfaces has only been possible with the inventions of the surface force apparatus (SFA) and the atomic force microscope (AFM) that were developed in the 1970s and 1980s, respectively.

13.2 SURFACTANT BEHAVIOR IN SOLUTION

Surfactants are chemical compounds that are surface active or more prevalent at surfaces or interfaces than in bulk media due to their dual property nature. Molecules that are surface active in aqueous media consist of both hydrophobic and hydrophilic entities. The hydrophilic entity is polar in nature and gives the molecule its ability to dissolve in water. Most often the hydrophilic portion consists of an ionic functional group such as sulfate, sulfonate, carboxylate, or amine, or it contains hydrophilic segments such as ethylene oxide. The hydrophobic portion of the molecule consists typically of CH_2 groups that are usually connected in continuous alkyl chains consisting of 4-18 CH_2 groups and an end group of CH_3. Some surfactants contain a main alkyl chain section and up to three small branches of a smaller number of CH_2 groups and a CH_3 end group. An example of the structure of a surfactant is shown in Figure 13.1 for cetyl benzyl trimethyl ammonium chloride. The hydrophobic portion of the molecule is non-polar in nature and is more favorably associated with other non-polar

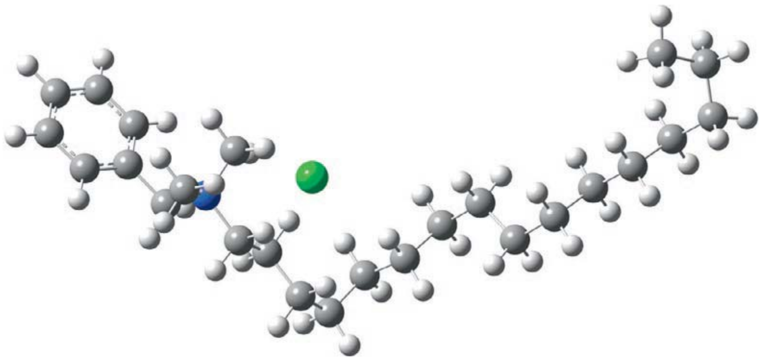

FIGURE 13.1 Molecular structure of cetyl benzyl trimethyl ammonium chloride for the lowest energy configuration with carbon, hydrogen, and nitrogen atoms, as well as a chloride ion, represented by gray, white, blue, and green spheres, respectively. *(Image obtained by Yakun Zhu.)*

entities such as air or other hydrocarbons, rather than water. Consequently, the dual nature of surfactant molecules is more readily accommodated at water-air or water-oil interfaces than in bulk water. Moreover, surfactant molecules are found in higher concentrations at interfaces than in bulk aqueous media because the hydrophilic portion of the molecule can associate with water molecules at the same time as the hydrophobic portion associates with non-polar media or hydrocarbon segments from other surfactant molecules. The tendency for surfactant molecules to associate with interfaces and other surfactant molecules leads to a high tendency for adsorption and aggregation, which increases as the hydrocarbon chain length increases and/or when the polarity of the solvent increases.

As surfactant concentration increases, the tendency of the surfactant to adsorb and/or form aggregate structures increases. At low concentrations, surfactant molecules do not have sufficient energy to form aggregates or adsorb at high concentrations at interfaces, as illustrated in Figure 13.2, although they adsorb at higher concentrations at interfaces than in bulk media. At moderate concentration levels, surfactant molecules also adsorb at interfaces at higher levels than in the bulk solution.

The accumulation of surfactant molecules at interfaces such as the air-water interface results in a lowering of the surface tension. In other words, the accumulation of surfactant molecules at interfaces lowers the interfacial energy due to the fact that the surfactant molecules have both hydrophilic and hydrophobic sections that are attracted to the water (polar) and air (non-polar) phases respectively.

Surface tension is often measured using a balance and a suspended hydrophilic object such as a plate or a ring. As shown in Figure 13.3, the presence of surfactant molecules at the air-water interface lowers the tendency of the water to climb up a hydrophilic substrate such as the plate shown in the figure, resulting in a reduced downward force on the object, as indicated by the arrows in

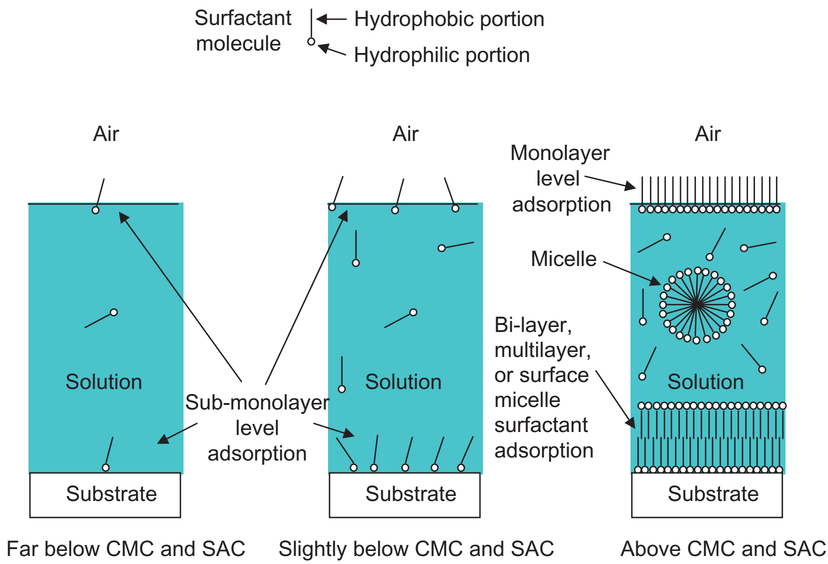

FIGURE 13.2 Schematic diagrams illustrating surfactant aggregation and adsorption as a function of relative surfactant concentration.

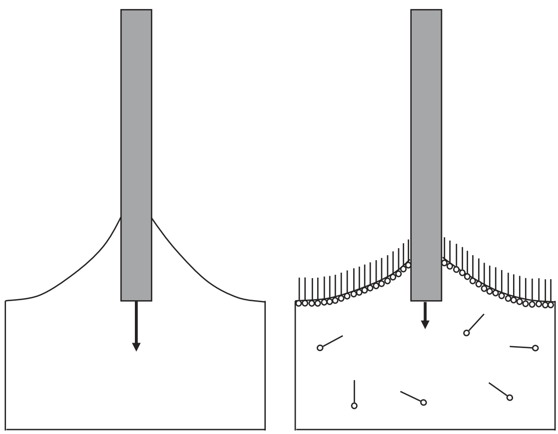

FIGURE 13.3 Schematic comparison of surface tension force exerted by water on a plate with (right side) and without (left side) surfactant present.

Figure 13.3. A typical plot of surface tension versus the natural logarithm of the concentration is shown in Figure 13.4. Note that the surface tension decreases rapidly with increasing surfactant concentration above a minimum concentration level. At higher concentrations it reaches a minimum value at a specific concentration level that is known as the critical micelle concentration (CMC) that will be discussed in more detail later.

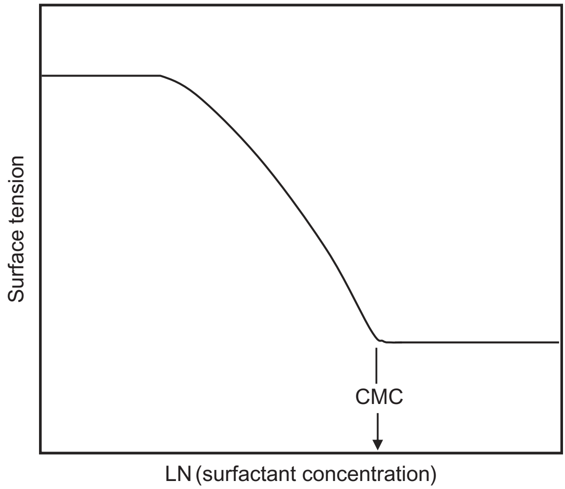

FIGURE 13.4 Sketch of typical surface tension versus ln(surfactant concentration) plot showing the critical micelle concentration (CMC).

The excess concentration of surfactant at an interface can be quantified using surface tension data and the Gibbs equation. The surface excess or concentration at the interface that is in excess of the bulk concentration can be expressed as[37]

$$\Gamma = -\frac{1}{RT}\left(\frac{d\gamma}{d\ \ln(c)}\right) \tag{13.1}$$

below the CMC for constant-temperature tests in which no phase transformations occur, where R is the gas constant, T is the absolute temperature, γ is the surface tension, and c is the concentration of surfactant. Note that the slope of the surface tension plot shown in Figure 13.4 is used in the calculation of the surface excess, which is also nearly equivalent to the adsorption density.

The adsorption of surfactant molecules at interfaces is often evaluated using the Langmuir model, which begins with the equilibrium adsorption reaction:

$$A + S = AS \tag{13.2}$$

in which A is the adsorbate, S is the surface site available for adsorption, and AS represents a surface site on which the adsorbate, A, is adsorbed. Using a simple equilibrium expression, the equilibrium constant K'_A can be expressed as

$$K'_A = \frac{C_{AS}}{C_A C_S} \tag{13.3}$$

in which C_{AS} is the surface concentration of adsorbed adsorbate, C_S is the surface concentration of unoccupied surface adsorption sites, and C_A is the bulk solution concentration of adsorbate. Using Equation (13.3) with a surface site balance and substitution for the concentration of available sites ($C_S = C_{tot} - C_{AS}$) leads to a common expression for the fraction of surface covered by the adsorbate:

$$\theta = \frac{C_{AS}}{C_{\text{Total sites}}} = \frac{K'_A C_A}{1 + K'_A C_A} \tag{13.4}$$

However, it should be noted that this expression is only valid at concentration levels below those needed for surfactant aggregation. It is also important to realize that surfactant aggregation at solid surfaces, which is referred to as the surface aggregation concentration (SAC), often occurs at concentrations that are below the CMC. Thus, the CMC applies to surfactant micelle aggregate formation in solution and the SAC applies to surfactant aggregate formation on solid surfaces.

As the surfactant concentration reaches levels that exceed the SAC, the surfactant molecules have already exhausted interfacial adsorption sites at the monolayer level, and the free energy of the system warrants additional association and aggregation in the form of spherical or cylindrical micelles as well as bilayers and/or multilayers at interfaces. Similarly above the CMC, micelles form in solution. Micelles are spherical or cylindrical aggregates of surfactant molecules in which the hydrocarbon portions of the molecules associate with the hydrocarbon portions of other surfactant molecules in the interior of the structure. The outer wall of the micelle structure is formed with the hydrophilic functional groups of the surfactant molecules as depicted in Figure 13.2, which illustrates surfactant adsorption relative to particles and surfaces.

The tendency for surfactant molecules to aggregate and associate at interfaces is influenced strongly by the length of the hydrocarbon portion of the surfactant molecules. As the hydrocarbon chain portion of the surfactant molecule increases in length, the tendency of the surfactant to aggregate or adsorb increases to relieve energetically unfavorable interactions between polar solvent molecules and non-polar hydrocarbon groups. Consequently, the critical concentration for aggregation (CMC or SAC) decreases as the chain length increases because the aggregation event occurs at decreasing concentration levels as the hydrocarbon chain length increases. The effect of increasing hydrocarbon chain length on the aggregation concentration (CMC for solutions) can be observed readily from the data in Table 13.1. The data in Table 13.1 indicate that the CMC of alkyl trimethyl ammonium chloride surfactants decreases by approximately a factor of 3 for every two CH_2 units added to the surfactant molecule.

The tendency for surfactant molecules to adsorb and/or aggregate in solution is also influenced by the solution environment. As the solution becomes more polar, the tendency for surfactant molecules to aggregate in solution and/or

TABLE 13.1 Comparison of Hydrocarbon Chain Length and CMC for Alkyl Trimethyl Ammonium Chlorides in Water[34]

Number of CH_2 Segments	CMC (M)
10	6.3×10^{-2}
12	1.9×10^{-2}
14	4.5×10^{-3}
16	1.3×10^{-3}
18	3.4×10^{-4}

TABLE 13.2 Comparison of Surfactant CMC Values as a Function of Ionic Strength and Surfactant Type

Surfactant	Ionic Strength	Chain Length (Number of CH_2 Segments)	CMC (M)
Sodium octyl sulfate	0.3	8	6.7×10^{-2}
Sodium octyl sulfate	0.07	8	1.3×10^{-1}
Sodium decyl sulfate	0.3	10	6.9×10^{-3}
Sodium decyl sulfate	0.007	10	3.3×10^{-2}
Sodium dodecyl sulfate	0.3	12	7×10^{-4}
Sodium dodecyl sulfate	0.0007	12	8.1×10^{-3}
Cetyl pyridinium chloride	1.0	16	3×10^{-6}
Cetyl pyridinium chloride	0.0003	16	3×10^{-4}

Data were taken from Refs. 34 and 36.

adsorb at interfaces increases. One factor that quantifies the polar nature of the solvent is the ionic strength. Consequently, the effect of ionic strength on surfactant aggregation is very significant, as evidenced by the CMC values given in Table 13.2. Note that the CMC values in Table 13.2 are considerably lower in high ionic strength media. The data in Table 13.2 also indicate that the effect of ionic strength is more pronounced as the hydrocarbon chain length increases, indicating that the surfactant molecules with longer alkyl hydrocarbon chains have an increased tendency to leave the solution to form surface and solution aggregates as the solvent polarity increases.

The combined effects of environment and hydrocarbon chain length on the CMC of a surfactant can be predicted based on a combination of theory and empirical information. The following equation is very useful in predicting the CMC (or SAC if slightly different values for x, $\Delta G_{\mathrm{c.l.}}$, and k are used) of surfactant molecules under a variety of chain length and ionic strength scenarios[38]:

$$\mathrm{CMC} \cong \exp\left(\frac{1}{RT}[(L-x)\Delta G_{\mathrm{c.l.}} + k(L-x)RT\ \ln(\gamma_{\mathrm{m}})]\right) \tag{13.5}$$

in which R is the gas constant, T is the absolute temperature, L is the total number of consecutive CH_2 units in the surfactant molecule, $\Delta G_{\mathrm{c.l.}}$ is the free energy increment for each CH_2 unit (−1500 J/mol is a good estimate for some surfactants[38]), x is the number of CH_2 units in the surfactant molecule necessary to initiate surface activity (usually around 5), k is a solvent polarity factor (usually 1.0-1.5 for surfactants for which L is 12-18[38]), and γ_{m} is the activity coefficient determined using an ionic activity coefficient equation, such as the Davies equation, which can be expressed as[39]:

$$\gamma_{\mathrm{m}} = 10^{-0.5083z^2}\left(\frac{\sqrt{I}}{1+\sqrt{I}} - 0.2I\right) \tag{13.6}$$

at 25 °C, where z is the charge of the surfactant ion (usually 1) and I is the ionic strength[39]:

$$I = 0.5\sum_{j=1}^{n} m_j z_j^2 \tag{13.7}$$

where $j=1$ to n represents all positively and negatively charged species in solution and m is the molality. More exact values of the various constants can be obtained using CMC data based on calibration plots as described by Free et al.[38]. More advanced models for CMC prediction and related salt and ionic strength effects are available in the literature,[40–49] but they generally require more information for their application.

One example of more advanced modeling for surfactant aggregation involvesthe use of molecular dynamics simulations. Molecular dynamics simulations utilize basic laws of physics and atomic level interaction forces to predict movements of molecules in a variety of environments. An example of a molecular dynamics simulation result for cetyl benzyl trimethyl ammonium chloride surfactant aggregation and adsorption on an iron surface is shown in Figure 13.5. Related molecular dynamics simulation information is found in the literature.[50–54]

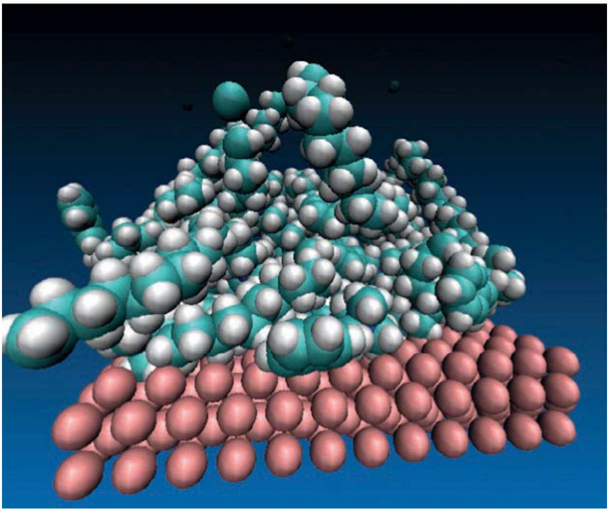

FIGURE 13.5 Molecular dynamics simulation of cetyl benzyl trimethyl ammonium chloride ion aggregation and adsorption at an iron surface. The red, white, green, and blue spheres represent iron, hydrogen, carbon, and nitrogen, respectively. Chloride ions and water molecules were present in the simulation but are not shown. *(Image courtesy of Dr. Keith Prisbrey.)*

Understanding how surfactants behave in solution in terms of their tendency to aggregate and adsorb at interfaces is a critical prerequisite to understanding how they can enhance particle removal, as will be demonstrated in a subsequent section.

13.3 INTERFACIAL FORCES AND PARTICLE REMOVAL

13.3.1 Introduction to Interfacial Forces

Interaction forces between molecules and surfaces can be very powerful and influential. The forces between molecules determine the physical state of groups of molecules. Such groups of molecules make up surfaces and become the origin of the forces that exist between surfaces. The forces between molecules and surfaces in aqueous media are often separated into electrostatic, van der Waals, structural, and hydrophobic forces. In situations involving non-immersion scenarios, capillary forces must also be considered, although in the remainder of this chapter they will be neglected.

van der Waals forces arise from the interaction of atomic and/or molecular dipole interactions.[40] van der Waals forces between molecules or atoms are attractive in nature, and such attractive forces exist even when the atom or molecule does not have a permanent dipole, since the dipole can be induced.[55] The van der Waals force between a sphere and a plate can be expressed as[55]

$$F(h) = -\frac{Ar}{6h^2} \tag{13.8}$$

where A is the Hamaker constant, r is the radius of curvature of the particle, and h is the separation distance between the particle and the plate.

The electrostatic force between a sphere and a plate is given as[55]

$$F(h) = \frac{128\pi R\rho_\infty kT\gamma_1\gamma_2}{\chi}\exp(-\chi h) \tag{13.9}$$

where the surface charge for each surface (1 and 2 as denoted by subscripts) is defined as[55]

$$\gamma_{(1\text{ or }2)} = \tanh\left(\frac{ze\Psi_{0\,(1\text{ or }2)}}{4kT}\right) \tag{13.10}$$

and the Debye length, χ, is given as[53]

$$\chi = \left(\frac{2000e^2IN_\mathrm{A}}{\varepsilon_0\varepsilon_\mathrm{m}kT}\right)^{1/2} \tag{13.11}$$

in which z is the ion valence, e is the electrical charge, ρ_∞ is the bulk dissolved ion density, $\Psi_{0\,(1\text{ or }2)}$ is the surface potential for the surface of interest (surface 1 or 2), I is the ionic strength ($0.5\Sigma z^2\rho_\infty$ for all positive and negative ions), N_A is Avogadro's number, ε_0 is the dielectric permittivity of vacuum, and ε_m is the dielectric constant of the medium between the two surfaces.

In many aqueous environments involving hydrophilic surfaces, the forces between surfaces are dominated by the electrostatic and van der Waals forces, since structural forces tend to be oscillatory in nature and tend to average to a near-zero net effect. In other words, the structural forces show up as oscillations above and below the other net forces. The trend of the other forces remains dominant despite the local oscillations that arise as surfaces come close to contact and individual layers of water are removed or added in a stepwise manner, causing stepwise oscillations in the overall force. Such structural oscillations are often observed within a few nanometers of the surface.[36]

The use of the combination of electrostatic and van der Waals forces to describe surface interactions was demonstrated extensively by Derjaguin, Landau, Verwey, and Overbeek. Their work and use of surface force theory has become widely known as the DLVO theory.[55] Furthermore, when only the electrostatic and van der Waals forces are considered in a surface force analysis, the resulting interaction force is sometimes referred to as the DLVO force. In situations involving hydrophilic substrates in low ionic strength media, the DLVO theory is often adequate to describe the forces between particles and surfaces before contact occurs.

Although the DLVO theory accounts for surface forces in many situations, it does not describe the effect of forces between hydrophobic surfaces that can far exceed the attractive forces predicted by the DLVO theory. The strong, attractive forces between hydrophobic surfaces are often referred to as hydrophobic forces.[55–58] There is considerable debate regarding the origin of these forces.[55] However, the most commonly accepted hypotheses suggest that the forces are related to water structuring effects and/or gas entrainment/cavitation effects.[55]

13.3.2 Measurement of Surface Forces

Surface forces can be measured by different instruments. Most commonly, surface forces are measured using the AFM and the SFA, both of which are capable of measuring interaction forces between surfaces at varying separation distances. The AFM, which was developed in the mid-1980s by Binnig and coworkers,[59] utilizes a thin cantilever beam that bends in response to surface forces experienced by the tip as the surface is moved closer to it by a piezocrystal (see Figure 13.6). The change in the angle of the cantilever causes a change in the angle of the light that is reflected off the tip of the cantilever (see Figure 13.6). Because the light that is reflected off the cantilever is measured by a set of detectors with respect to both quantity and position, a determination of the force can then be made based on the cantilever properties and dimensions combined with the position and intensity of the deflected light. The distance is measured based on the applied piezocrystal voltage change combined with the properties of the piezocrystal actuator and a calibration procedure.

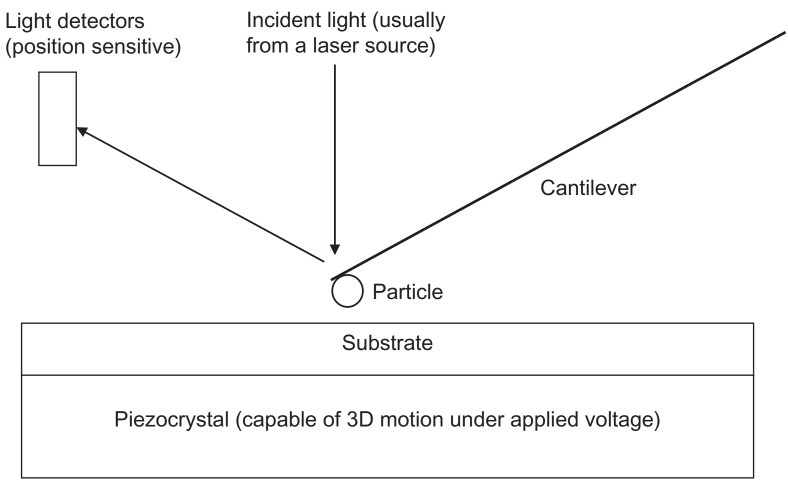

FIGURE 13.6 Schematic diagram of an atomic force microscope (AFM).

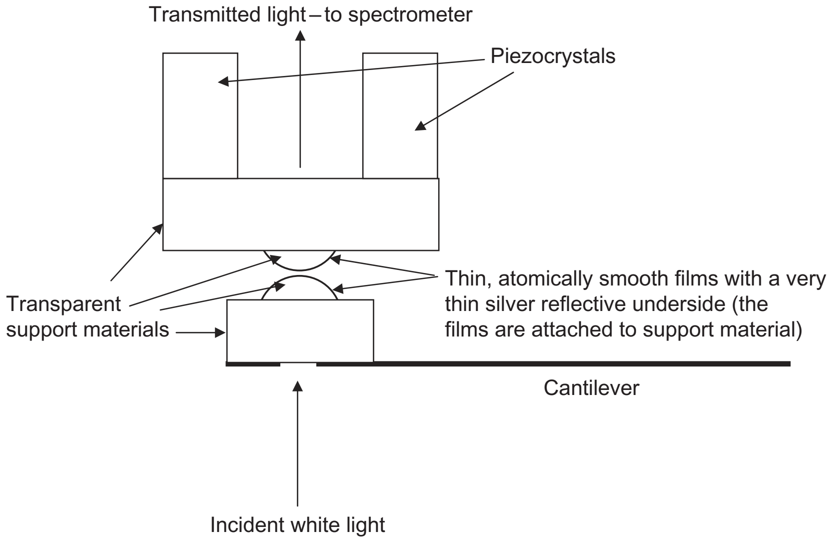

FIGURE 13.7 Schematic diagram of a surface force apparatus (SFA).

The SFA, which was developed by Tabor and coworkers prior to the AFM,[36] has a significantly different design than the AFM (see Figure 13.7). The SFA consists of a cantilever to measure the force based on deflection, as is done in the AFM. However, the separation between surfaces is measured using white-light diffraction techniques that utilize a spectrometer to analyze the various wavelengths of light. The surfaces that are used in the SFA must be atomically smooth, and they are generally mounted as thin films onto cylindrical, optically transparent support structures, as shown in Figure 13.7. The films must be coated with a very thin layer of silver or other reflective metal on the back side to allow significant light transmission while also allowing significant reflected light. As light reflects between the silvered back sides of the layers that approach contact, interference fringe patterns are created. The interference fringe patterns can be analyzed spectroscopically to determine the separation distance between surfaces to within approximately 1 Å.[36]

13.3.3 Adhesion

Particles adhere to surfaces due to favorable interaction forces between them. The dominant force of attraction between particles and surfaces is the van der Waals force. This force is so strong that the particle and surface are deformed upon natural contact. The extent of the deformation is determined by the elastic moduli of the substances involved, as well as the magnitude of the attractive force. The deformation is large for soft materials and small for hard materials. Several theories have been proposed to calculate the deformation and forces involved in the adhesion of particles to surfaces.[60–65] The attractive forces remain the same despite the material deformation, but the effect of material deformation leads to an increased requirement for particle removal.[36]

Adhesion forces can be estimated from the sum of the known forces. The force calculation is usually made at a particle-to-surface separation distance that is between 0.1 and 0.5 nm in aqueous media. The separation distance is not zero because the continuum theories used to derive the force expressions do not accurately account for the discrete nature of the atoms at the interfaces at close separation distances. Also, in aqueous media, surface layers of water are bonded to the surfaces and may affect the effective separation distance. Another approach, the surface energy/acid-base theory,[66] also provides useful approximations of adhesion forces, although this technique will not be discussed further because it cannot be applied as easily to systems involving surfactant adsorption as the sum-of-forces approach described previously. In addition, the surface energy/acid-base theory does not account for electrostatic forces that can be important in particle-surface systems.

13.3.4 Particle Removal Forces

Particles can be removed from surfaces by pulling, rolling, or sliding them off, yet each of these techniques requires an external force. The most common external force that is used for these techniques is fluid drag that is created by fluid flow. The force of fluid drag on a particle under laminar flow conditions can be expressed as[67]

$$F = -3\pi d_0 V \mu \tag{13.12}$$

where F is the force, d_0 is the particle diameter, V is the fluid velocity, and μ is the viscosity. However, the fluid velocity at any given position relative to the particle is proportional to distance above the substrate surface.[68] Consequently, the fluid elements that are capable of moving the particle are located at a distance from the surface that is proportional to the particle diameter. Thus, as is intuitively apparent, the force of removal due to fluid drag is surface area dependent based on fluid flow near a substrate surface and an attached particle at the surface. Consequently, the actual removal force on a particle at a surface due to fluid drag is directly proportional to the diameter squared. Thus, as particles become smaller, the force available for removal from fluid movement decreases rapidly. In contrast, the forces such as van der Waals and electrostatic forces that control adhesion are linearly related to particle diameter and do not diminish as rapidly as the traditional removal forces. Therefore, large particles are easily washed away from surfaces, and small particles are not easily washed away from surfaces unless the surface forces are reduced or the removal forces enhanced, although in this chapter only surface force modification using surfactants will be discussed as a means of enhancing particle removal.

13.3.5 Modification of Surface Forces Using Surfactants

The adsorption of surfactant molecules at the surfaces of particles and substrates can alter the van der Waals attractive force between the particles and substrates as well as the electrostatic, structural, and hydrophobic forces.

The effect of surfactant adsorption on the van der Waals force is most easily quantified using an approximate expression developed by Israelachvili[36] (for surfaces with adsorbed layers that are separated by a medium such as water) that has been modified from a plate-plate interaction force to a plate-sphere interaction force using the Derjaguin approximation[56]:

$$F(h) = \frac{R}{6}\left[\frac{A_{432}}{h^2} - \frac{\sqrt{A_{545}A_{323}}}{\left(h+t_p\right)^2} - \frac{\sqrt{A_{121}A_{343}}}{\left(h+t_s\right)^2} + \frac{\sqrt{A_{121}A_{545}}}{\left(h+t_p+t_s\right)^2}\right] \tag{13.13}$$

where 1 is the surface of the substrate, 2 is the coating on the substrate surface, 3 is the medium separating the surfaces, 4 is the coating on the particle, 5 is the surface of the particle, t_s is the thickness of the coating on the substrate surface, and t_p is the thickness of the coating on the particle surface. However, in order to utilize this force expression, it is necessary to know the Hamaker constants between the various substances. A useful expression for determining the Hamaker constant based on Lifshitz theory is given by Israelachvili[36] as

$$\begin{aligned} A_{132} \approx & \frac{3}{4}kT\left(\frac{\varepsilon_1-\varepsilon_3}{\varepsilon_1+\varepsilon_3}\right)\left(\frac{\varepsilon_2-\varepsilon_3}{\varepsilon_2+\varepsilon_3}\right) \\ & + \frac{3h\nu_e}{8\sqrt{2}}\left[\frac{\left(n_1^2-n_3^2\right)\left(n_2^2-n_3^2\right)}{\left(n_1^2+n_3^2\right)^{1/2}\left(n_2^2+n_3^2\right)^{1/2}\left[\left(n_1^2+n_3^2\right)^{1/2}+\left(n_2^2+n_3^2\right)^{1/2}\right]}\right] \end{aligned} \tag{13.14}$$

where A_{132} is the non-retarded Hamaker constant between surfaces 1 and 2 across medium 3, k is the Boltzmann constant, T is the absolute temperature, ε is the dielectric constant of the specified medium, h is Planck's constant, ν_e is the average absorption frequency, and n is the refractive index. The justification for using a multiple-layer model to determine the effect of surfactant adsorption on the van der Waals force is made on the basis of experimental observation, which shows the effect of the adsorbed layer on the van der Waals force becomes increasingly important as the surfaces approach contact.[36] The adsorbed layer(s) of surfactant molecules also provides a steric barrier to direct contact between the particle and the substrate.

The adsorption of surfactant molecules on the substrate and particle surfaces also tends to significantly alter the charges on the surfaces, as is depicted with an adsorbed cationic surfactant molecule at a charged surface in Figure 13.8. The effect of surfactant adsorption depends on the existing charge prior to surfactant adsorption, the charge of the surfactant molecule, and the surface concentration of surfactant on each surface after adsorption. In some cases, the adsorption of ionic surfactant can have a very substantial effect on the electrostatic force

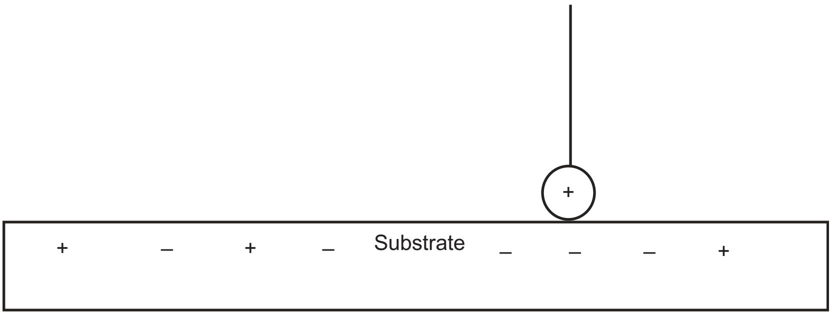

FIGURE 13.8 Schematic diagram of a cationic surfactant molecule adsorbed onto a charged substrate. Note that the charge on the substrate is generally due to adsorbed ions such as OH^- and H^+ that are not shown. Also note that water molecules, which play an important role at interfaces in aqueous media, are not shown.

between the substrate and particles in solution. In some particle removal applications, the particles have a charge that is opposite to that of the substrate, making the natural electrostatic force between the surfaces an attractive one (negative force). If an ionic surfactant adsorbs on the surfaces of the particles and the substrate, which is generally observed, both types of surfaces will likely develop the same charge as the surfactant, making the electrostatic interaction force between the surfaces a positive or repulsive force, thereby reducing particle adhesion and enhancing particle removal. In cases involving substrates and particles with similar natural charge, the adsorption of properly selected ionic surfactant molecules can enhance the existing repulsive electrostatic force. Thus, in most cases the adsorption of ionic surfactant molecules facilitates particle removal by providing or enhancing electrostatic repulsive forces in addition to providing a steric barrier to direct contact between the particle and the substrate.

Although it is clear that surfactant adsorption can reduce van der Waals attractive forces and create or enhance electrostatic repulsion between particles and surfaces, surfactant adsorption can lead to a hydrophobic attractive force, which tends to make particle removal more difficult. Because particle removal is the objective in this chapter, it is desired that the adsorbed surfactant molecules be arranged with hydrophilic ends projecting into aqueous media or hydrophobic ends projecting into a hydrophobic solvent. Using the illustrations shown in Figure 13.9, it is easily observed that for aqueous media and naturally hydrophilic surfaces this desired outcome is achieved only above the SAC where bilayers, multilayers, or surface micelles (spherical or cylindrical) form to provide an exposed interface of hydrophilic functional groups.

Surfactant adsorption and aggregation on naturally hydrophobic surfaces begins with adsorption of surfactant in the opposite orientation to that shown in Figure 13.9, with the hydrophobic tails attaching at the surface and the hydrophilic head groups extended into solution. Similarly, hemimicelles and hemicylindrical micelles adsorb on hydrophobic surfaces with the hydrophobic tails of

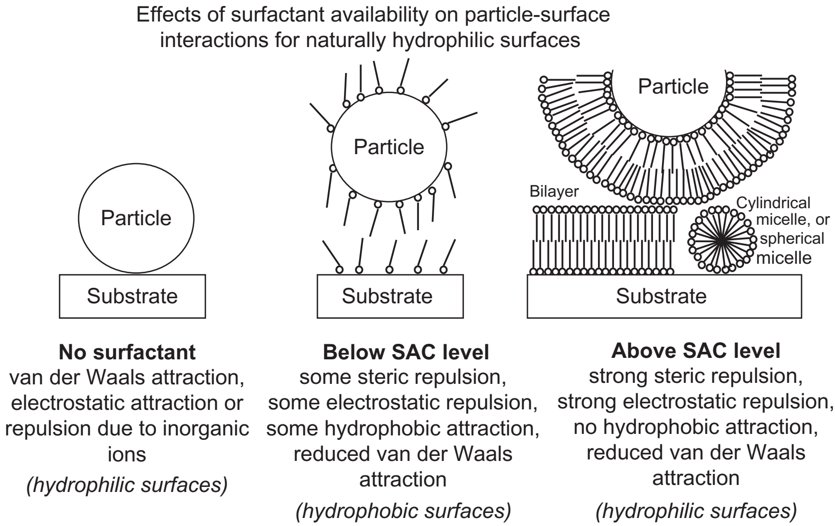

FIGURE 13.9 Schematic diagrams illustrating the effect of surfactant concentration and adsorption on the particle-substrate interaction for surfaces that are naturally hydrophilic. Note that water molecules and other ions such as counter ions have been omitted to simplify the drawings. Although the diagram shows bilayer formation above the SAC, it is also common to observe surface micelles and multilayer level adsorption above the SAC.

the surfactant molecules attached to the surface. Hydrophobic surfaces in aqueous media can achieve some level of removal when sufficient surfactant is present due to the presence of hydrophilic, charged surfactant head groups that extend away from the surface, contributing to hydrophilicity and charging, while also mitigating hydrophobic attraction to other hydrophobic surfaces.

Figure 13.10 shows an AFM image of cetyl trimethyl ammonium chloride surfactant adsorbed on a hydrophobic pyrolytic graphite surface in rows of hemicylindrical micelles. The corresponding width of each row of hemicylindrical micelles is approximately 5 nm, which corresponds approximately with the length of two surfactant molecules plus additional row separation due to electrostatic repulsion between functional groups.

The effect of the orientation of the head group and hydrocarbon tail of the adsorbed surfactant molecules on the attractive hydrophobic force has been demonstrated by Freitas and Sharma,[69] who showed a strong attractive force between hydrophilic silica particles and substrates below the SAC, where hydrocarbon tails of adsorbed surfactant molecules extended into solution to produce a hydrophobic interface. However, above the SAC, where only functional groups from a bilayer, multilayers, or surface micelles extended into aqueous media, the interaction force became very weak. It should be noted that no complete explanation for such hydrophobic interaction forces has been developed.

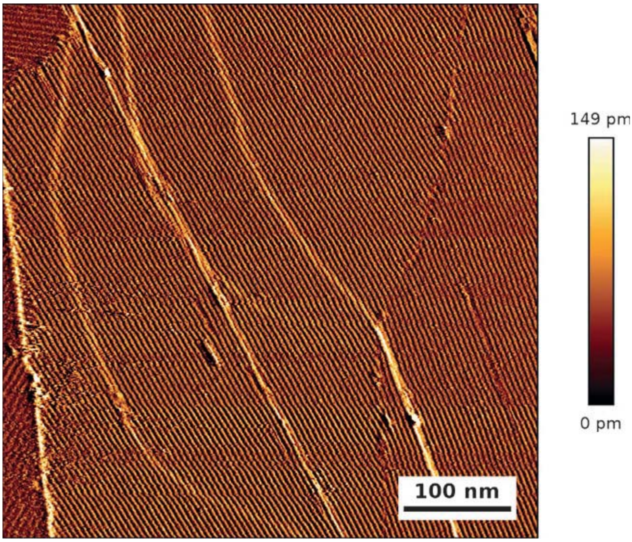

FIGURE 13.10 JPK Nanowizard 3a Ultra AFM image of cetyl trimethyl ammonium chloride surfactant adsorbed as hemicylindrical micelles on a highly oriented hydrophobic pyrolytic graphite surface. (Courtesy of JPK Instruments)

The net force between a particle and a substrate can be calculated using the equations described previously for the van der Waals and electrostatic force components, which are the most important overall forces as long as the surfaces are hydrophilic. Many surface force studies have been performed,[70–84] and there is at least general agreement that experimental data can be modeled using the applicable theory.[74,84] Thus, it is useful to examine the effects of surfactant adsorption on the net interaction force between a particle and a substrate. There is significant interest in understanding particle-substrate interaction forces in the context of removing particles from surfaces that is of particular interest in the electronics manufacturing industry, particularly in association with the process of chemical mechanical planarization and related cleaning of wafers.[85–90]

A sample comparison of DLVO forces that have been normalized by dividing by particle radius versus the separation distance between bare and film-covered surfaces is presented in Figure 13.11 for an ideal alumina sphere that is approaching a quartz surface in aqueous media at a near-neutral pH. Note that the net normalized force is highly negative (attractive) as the particle approaches the surface when no surfactant is present. The attraction in the absence of surfactant is due to the van der Waals attractive force and an attractive electrostatic force. However, in the presence of cetyl pyridinium chloride (CPC) surfactant at a level of 0.0028 M, which is well above the CMC of 0.0003 M (also above the SAC of around 0.0001 M) for this solution, it results

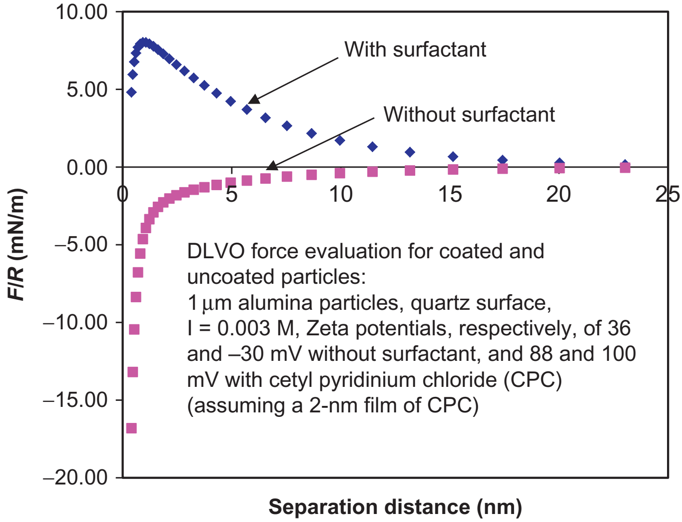

FIGURE 13.11 Comparison of normalized DLVO force (normalized by dividing by particle radius) versus separation distance for alumina particles approaching a quartz surface. Note that the Hamaker constant was calculated using Equation (13.14) with refractive index and dielectric constant data from Ref. 36. The forces were calculated using Equations (13.9)–(13.11), (13.13), and (13.14). Note that the zeta potential values shown in the figure are values that were measured using ground quartz and alumina particles in a 0.003 M KCl solution at 22 °C with and without 0.0028 M cetyl pyridinium chloride as indicated. The surface potentials, ψ, were calculated based on the zeta potentials using the relationship: $\psi = \text{zeta potential}\ [\exp(-\chi/1\ \text{nm})]$ (assumes that the zeta potential is 1 nm from the surface).

in a positive or repulsive force at separation distances that are representative of contact. It should be noted that in applying the DLVO theory in aqueous media, contact is assumed to occur at a separation distance greater than zero (often around 0.4 nm), as explained previously. In this particular case, the result of adding surfactant is quite dramatic because the surfaces had opposite charges prior to surfactant adsorption. Thus, the addition of the surfactant made an enormous difference to the surface forces. In systems involving similar surfaces, surfactant molecules do not alter the forces as dramatically, yet the addition of surfactant molecules can enhance repulsive forces as shown in Figure 13.12 for a quartz sphere approaching a quartz surface, thereby enhancing the potential for removal. Other studies also show that surfactants have a significant impact on particle-surface interaction forces.[80,91–93]

13.3.6 Measurement of Particle Removal

Particle removal effectiveness is nearly always measured by optical methods. One simple approach is to capture digital images of appropriately magnified sections of a surface that has residual particles. The digital images can then be

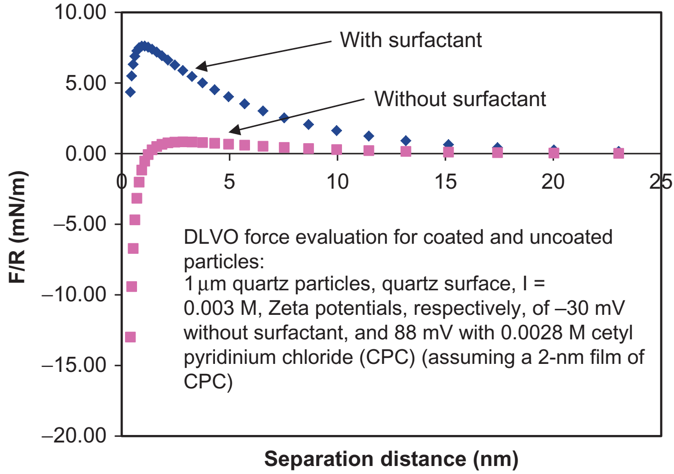

FIGURE 13.12 Comparison of normalized DLVO force (normalized by dividing by particle radius) versus separation distance for quartz particles approaching a quartz surface. Note that the Hamaker constant was calculated using Equation (13.14) with refractive index and dielectric constant data from Ref. 36. The forces were calculated using Equations (13.9)–(13.11), (13.13), and (13.14). Note that the zeta potential values shown in the figure are values that were measured using ground quartz particles in a 0.003 M KCl solution at 22 °C. The surface potentials were calculated based on the zeta potentials using the relationship: $\psi = $ zeta potential [exp $(-\chi/1$ nm)] (assumes that the zeta potential is 1 nm from the surface).

analyzed using image analysis software to determine the number of residual particles. A similar method that is often used in the electronics industry to inspect wafer contamination involves the use of laser scanning and associated detectors that measure light that is scattered by particles or defects on the surface.

13.3.7 Enhanced Particle Removal Results Associated with Surfactant Use

Surface force calculations show that surface forces can be reduced significantly by the presence of surfactant, and it is therefore anticipated that the addition of sufficient surfactant will enhance particle removal. Several studies show that surfactants can assist in reducing the number of residual particles on a surface.[41,53,58,94–98] Several sets of particle removal data are presented in Figures 13.13–13.18. Each figure shows that the presence of surfactant has a significant influence on the removal of particles from the associated substrates. Figures 13.13–13.15 and 13.18 show that the effectiveness of particle removal generally increases with increasing surfactant concentration. The observed removal effect associated with increasing surfactant concentration in Figure 13.18 is generally consistent with the theoretical force considerations

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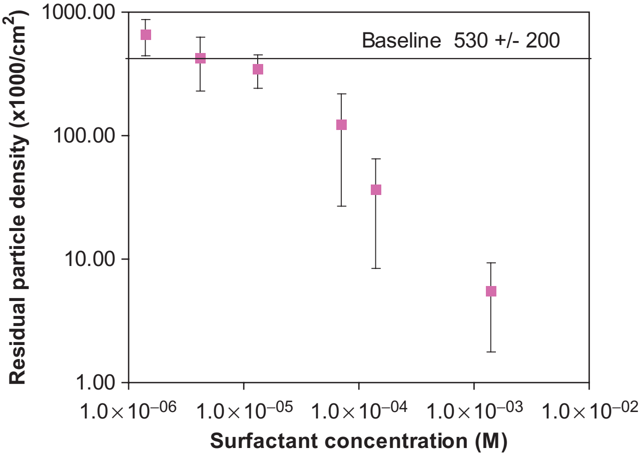

FIGURE 13.13 Residual particle density following CMP processing on copper in 1% ferric nitrate medium versus cetyl pyridinium chloride concentration at 31 °C. *(Data obtained and adapted from Ref. 96.)*

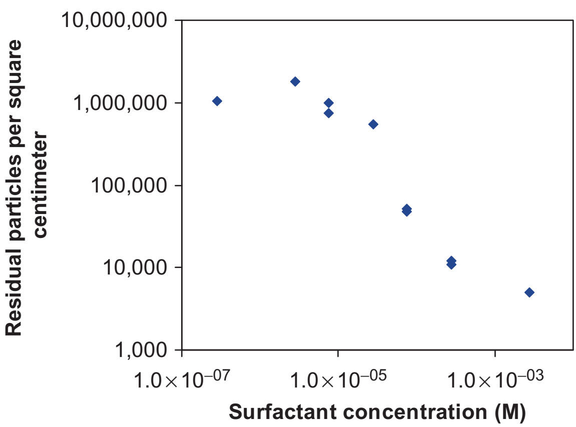

FIGURE 13.14 Comparison of residual particle density and cetyl pyridinium chloride (CPC) concentration following polishing of a tungsten-coated surface using 0.7-μm-diameter alumina particles.

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that were presented in Figures 13.11 and 13.12, which predicted an attractive force without surfactant and a repulsive force with sufficient surfactant.

Figure 13.18 shows that the effectiveness of the surfactant is significantly greater above the CMC. Because the CMC is generally significantly greater than the SAC due to higher interfacial energy at solid surfaces relative to the air-water interface, it is likely that the removal is largely ineffective below the SAC. However, Figure 13.16 shows that if the adsorption of surfactant onto a hydrophilic substrate leads to a hydrophobic surface as depicted in Figure 13.9

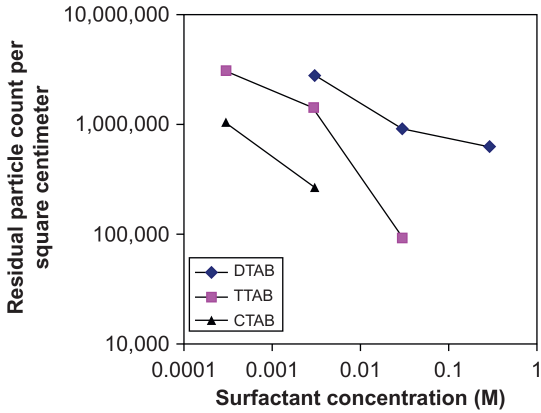

FIGURE 13.15 Comparison of residual particle density and surfactant concentration following polishing of a tungsten-coated surface using 0.7-μm-diameter alumina particles. DTAB, dodecyl trimethyl ammonium bromide; TTAB, tetradecyl trimethyl ammonium bromide; CTAB, cetyl trimethyl ammonium bromide.

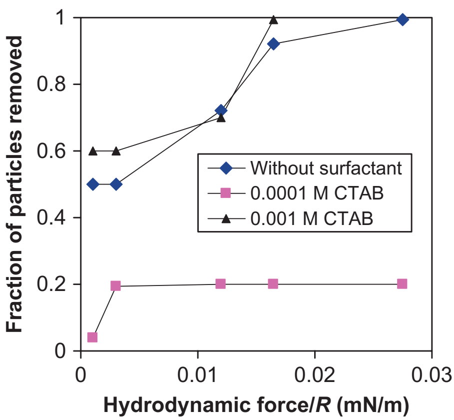

FIGURE 13.16 Effect of cetyl trimethyl ammonium bromide concentration on the removal of 10-μm hydrophilic glass particles from a hydrophilic oxidized silicon wafer substrate. The *x*-axis shows the hydrodynamic removal force due to fluid flow that is normalized by dividing by the particle radius to give a quantitative assessment of the relative removal force. *(Data adapted from Ref. 69.)*

through the adsorption of less than a bilayer or multilayer, removal is inhibited (see the data for 1×10^{-4} M surfactant, which is below the SAC). However, above the SAC, the surface becomes hydrophilic as depicted in Figure 13.9 and removal is facilitated (see the 1×10^{-3} M data set in Figure 13.16). These findings suggest that for hydrophilic substrate/particle systems the equilibrium solution surfactant concentration should exceed the SAC for maximum cleaning benefits.

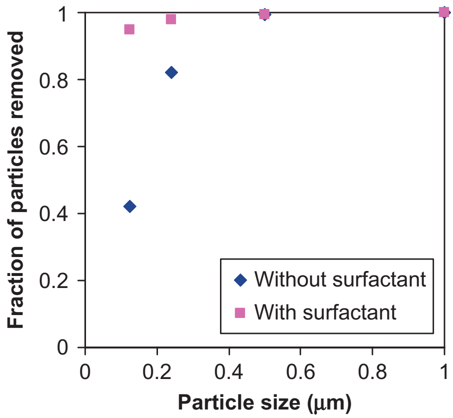

FIGURE 13.17 Polystyrene latex particle removal efficiency versus particle size from silicon substrates using 0.5% HF with and without 1% anionic surfactant. *(Data adapted from Ref. 98.)*

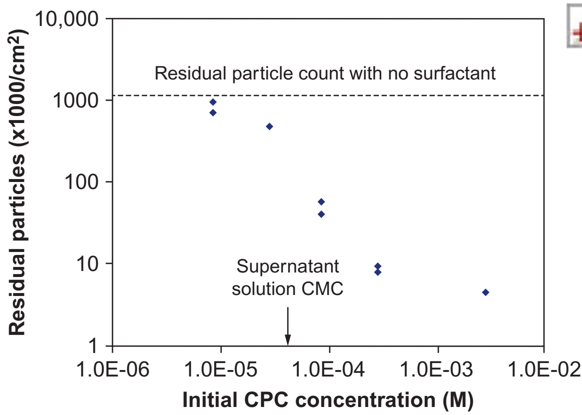

FIGURE 13.18 Comparison of residual particle density and cetyl pyridinium chloride (CPC) concentration following polishing of quartz glass using 0.7-μm-diameter alumina particles. Note that the CMC of the supernatant solution, which is based on the total CPC added to the solution, is much higher than the normal CMC (3×10^{-4} M) due to surfactant adsorption on alumina particles. In other words, the supernatant CMC indicates that the free, non-adsorbed surfactant concentration is close to the normal CMC even though the initial CPC concentration was much higher.

In contrast, however, the results shown in Figure 13.16 also lead to the inference that if the surfaces are naturally hydrophobic, enhanced removal will likely be encountered with surfactant at concentrations below the SAC. This inference can be explained on the basis of surface hydrophilicity. Because the surface will become more hydrophilic and charged with increasing surfactant adsorption below the SAC, the potential for hydrophobic attraction will be minimized and the opportunity for particle removal will be enhanced. It will also be enhanced above the SAC, since additional adsorption above the monolayer level occurs in paired layers or micelles, which always have hydrophilic exposed interfaces.

However, it should be understood that hydrophobicity/hydrophilicity cleaning predictions discount the effect of surface charging that is associated with surfactant adsorption as discussed in connection with electrostatic forces. Consequently, it is possible to observe enhanced particle removal when the surfaces have less than bilayer or multilayer coverage of surfactant molecules. However, due to the competing electrostatic repulsion and hydrophobic attraction associated with surfactant coverage below the multilayer level, particle removal is not likely to be as effective as it would be with coverage above the multilayer level, which occurs above the SAC.

Non-ionic surfactant molecules have been tested for their ability to assist in particle removal, but they tend not to perform as well as ionic surfactants.[94] This finding is not surprising when it is considered that the effect of enhanced favorable electrostatic forces can be generated by the adsorption of ionic surfactant but not non-ionic surfactant.

The overall surfactant cleaning results show that the effect of surfactant adsorption on surface charge, surface hydrophobicity/hydrophilicity, and steric interactions is significant within the context of enhancing particle removal. The results also highlight the importance of understanding the natural surface charge and hydrophobicity/hydrophilicity as well as the SAC, which are strong functions of ionic strength and surfactant properties, as discussed in Section 13.2. In comparing cleaning data sets it is also important to account for the difference between equilibrium and initial surfactant concentrations, which are often very different due to the high surface area associated with fine particles that can adsorb large quantities of surfactant and reduce the surfactant concentration in solution.

13.3.8 Post-Cleaning Surfactant Removal

After removing particles with the assistance of a surfactant, it is often necessary to remove the residual surfactant. Because most surfactant adsorption events are physical in nature, the surfactant can be removed, to some extent, by simply removing the surfactant from the surrounding solution in a rinsing operation. However, because adsorbed surfactant molecules bond with each other, they can resist removal during rinsing with pure water. Their resistance to removal is related to the number of carbons in the hydrocarbon chain. The longer the hydrocarbon segment, the more resistant they are to removal. The resistance to removal can be overcome to some extent through the addition of effective ions that compete for surface adsorption sites, form complexes, or react with the adsorbed surfactant. It has been shown that the exposure of CPC-coated tungsten surfaces to a solution of 1 M ammonium hydroxide for 5 min results in removal of the CPC, whereas simple rinsing in deionized water is only partially successful in removing the surfactant molecules.[99] The rare exception to easy removal with appropriate ions is the removal of chemisorbed surfactant molecules, which may require more aggressive removal environments.

13.3.9 Selection of Surfactants for Cleaning Purposes

The selection of surfactant molecules for enhanced particle removal from surfaces should involve the assessment of several important factors. One important selection criterion is the behavior of the surfactant molecule at the temperature of the application. Some surfactants have low solubility at the temperature of the desired application and should not be considered. Most often solubility is controlled predominantly by the number of hydrocarbon units in the alkyl chain, so chain length is an important parameter to consider with respect to solubility issues. Although increasing the chain length reduces solubility, it increases surfactant adsorption and aggregation tendencies. Consequently, the effect of chain length must be balanced between adsorption and solubility considerations.

Other factors that need to be considered are functional groups and the application environment. Anionic surfactants are not effective in highly acidic media because they form precipitates, and cationic surfactants are not effective in highly alkaline media due to precipitation. Zwitterionic surfactants are usually only effective in solutions with a near neutral pH and the absence of highly charged ions. Some environments that contain highly charged ions may induce precipitation of surfactant ions with opposite charges. Non-ionic surfactant molecules are not subject to the same precipitation concerns although their solubility can change significantly with temperature.

13.3.10 Mathematical Modeling of Enhanced Particle Removal Using Surfactants

Recent modeling efforts have shown that enhanced particle removal using surfactants can be predicted using an adsorption site inhibition or an energy-based approach.[100] The adsorption site inhibition approach consists of traditional Langmuir adsorption modeling with the assumption that surfactant coverage is not limited to surface adsorption sites on the surface of a solid substrate. Instead, it is asserted that adsorption sites are limited by the interaction of surfactant with the substrate and/or a surfactant-covered substrate. Consequently, the coverage used in the Langmuir adsorption model for the site inhibition approach is an effective coverage that is limited by a combination of factors such as interfacial charge, surface sites, and surfactant adsorption. Although this modeling approach tends to fit residual particle density data well for most typical data ranges, it does not predict a lower limit for enhanced particle removal.

The energy-based approach, which utilizes an energy-based equation that has a functional form that is analogous to the Arrhenius equation, has been shown to be a more effective approach to the modeling of enhanced particle removal using surfactants than the adsorption site inhibition model.[100] The energy-based model leads to the equation[100]

$$P = P_0 \exp\left[z\left(\frac{K(C/\mathrm{SAC})}{1+K(C/\mathrm{SAC})}\right)^n\right] \tag{13.15}$$

where P is the residual particle density with surfactant present, P_0 is the residual particle density without surfactant present, z is an energy-based constant, K is a surfactant adsorption constant, C is the surfactant concentration, SAC is the surface aggregation concentration, and n is a constant.

The value of P_0 can be obtained by direct measurement. A plot of $\ln|\ln(P_0/P)|$ versus the natural logarithm of $(1+KC/\mathrm{SAC})/(KC/\mathrm{SAC})$ can be used to calculate the other constants based on the rearranged form of the energy-based residual particle density modeling equation:

$$\ln\left|\ln\left(\frac{P_0}{P}\right)\right| = n\,\ln(z) + n\,\ln\left(\frac{1+K(C/\mathrm{SAC})}{K(C/\mathrm{SAC})}\right) \tag{13.16}$$

Thus, the slope of the resulting plot will be equal to n and the value of n, combined with the intercept of the plot, can be used to determine z. The value of SAC can be estimated using Equation (13.5). The value of K can be estimated using surface tension plots, corrosion inhibition data, or Langmuir-Blodgett film compression tests.[101] The results of the fit of the energy-based model to residual particle density data, which are presented in Figures 13.19 and 13.20, show that this modeling approach is effective in predicting the effect

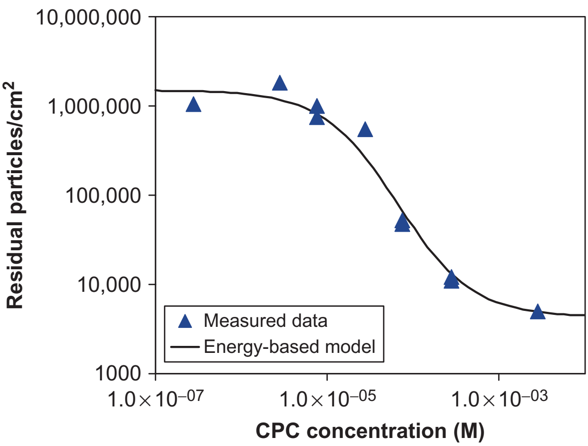

FIGURE 13.19 Comparison of residual particle density and cetyl pyridinium chloride (CPC) concentration following polishing of quartz glass using 0.7-μm-diameter alumina particles in 1% ferric nitrate medium at 31 °C. The energy-based model fit, which is shown as the solid line, used the following parameters: $P_0 = 1{,}500{,}000$ particles per cm^2, $z = 14{,}500$, $K/\mathrm{SAC} = 15{,}000$ l/mol, and $n = 0.99$. *(Data adapted from Ref. 100.)*

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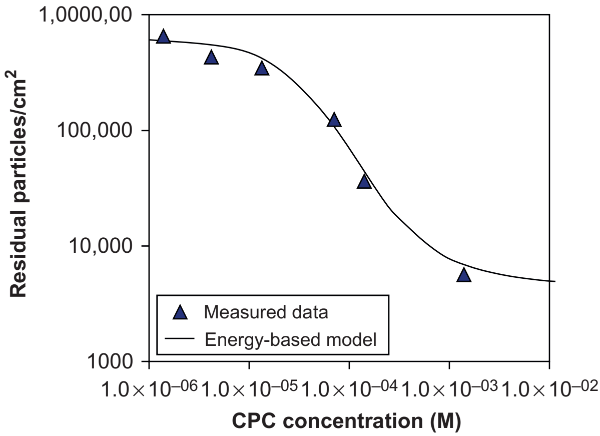

FIGURE 13.20 Comparison of residual particle density and cetyl pyridinium chloride (CPC) concentration following polishing using 0.7-μm-diameter alumina particles on copper in 1% ferric nitrate medium at 31 °C. The energy-based model fit, which is shown as the solid line, used the following parameters: $P_0 = 600{,}000$ particles per cm^2, $z = 12{,}000$, $K/\mathrm{SAC} = 11{,}500$ l/mol, and $n = 1.27$. *(Data adapted from Ref. 100.)*

of surfactant in enhancing alumina particle removal from both copper and quartz substrates.

13.4 SUMMARY

Surfactants can enhance particle removal from surfaces by modifying the particle-surface interaction forces. Adsorbed surfactant molecules can alter the van der Waals attractive force, electrostatic force, hydrophobic force, as well as provide a steric barrier to contact. The effect of surfactants on these forces can result in greatly enhanced particle removal efficiency.

Surfactant adsorption density and structure are important factors in determining removal enhancement performance associated with surfactants. Cleaning is generally most effective above the SAC, which for naturally hydrophilic surfaces allows for bilayer or multilayer level surfactant coverage that provides significant charge repulsion as well as a steric barrier. Adsorption below the monolayer level renders naturally hydrophilic substrates hydrophobic, which tends to reduce removal efficiency. In contrast, naturally hydrophobic surfaces are likely to benefit from both sub-monolayer and multilayer coverages of surfactant that occur, respectively, below and above the SAC. Existing adsorption theory and available formulas can aid in the prediction of the SAC, which is an important parameter in predicting the performance of surfactants in particle removal enhancement. Equations are also available to predict the effectiveness of surfactants in enhancing particle removal.

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